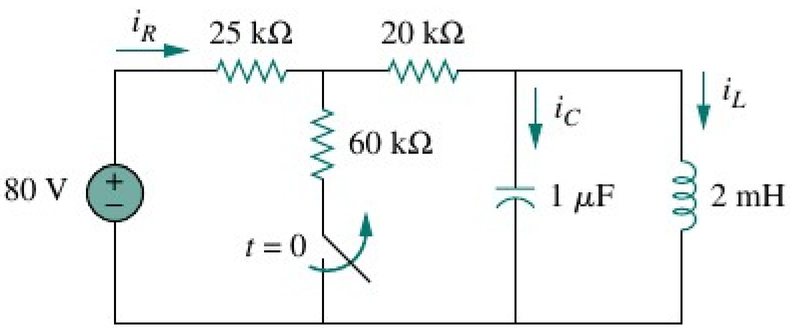
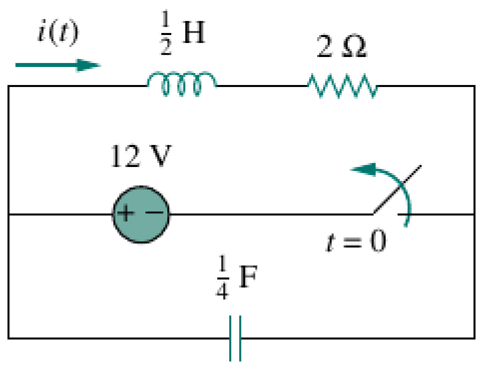
CSE 231 HW#2

RLC Circuits :

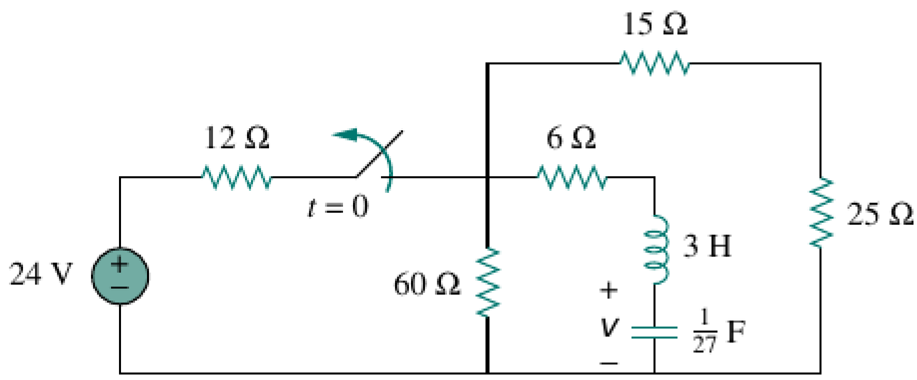
1) For the circuit given below :
   a) $i_L(0^+), v_C(0^+)$ and $v_R(0^+)$
   b) $di_R(0^+)/dt, di_L(0^+)/dt$ and $di_C(0^+)/dt$
   c) $i_R(\infty), i_L(\infty)$, and $i_C(\infty)$



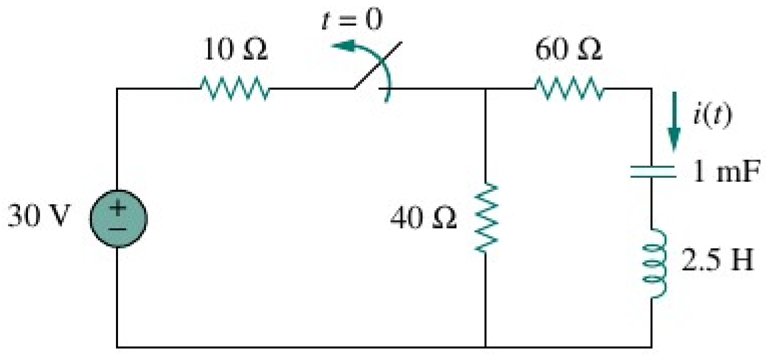
2) The switch in the circuit below has been closed for a long time but is opened at $t = 0$. Determine $i(t)$ for $t > 0$



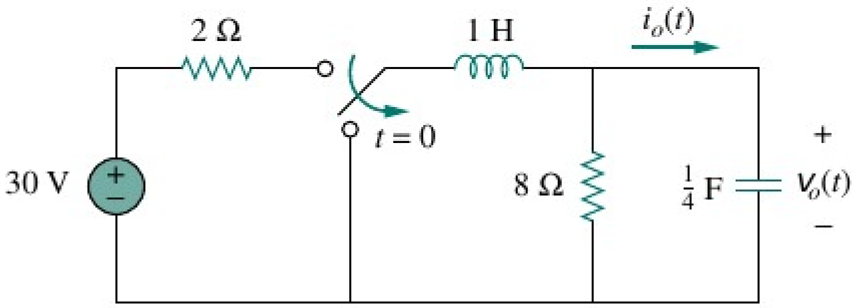
3) Calculate $v(t)$ for $t > 0$ for the circuit given below



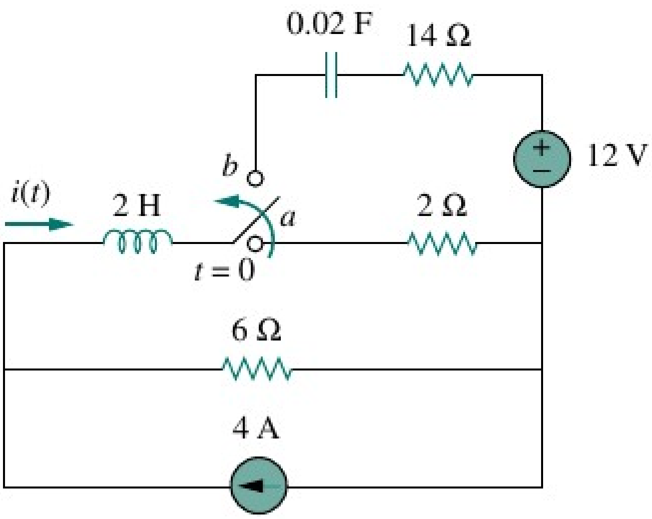
4) For $i(t)$ for $t > 0$ in the following circuit



5) Calculate $i_o(t)$ and $v_o(t)$ for $t > 0$ for the circuit given below :

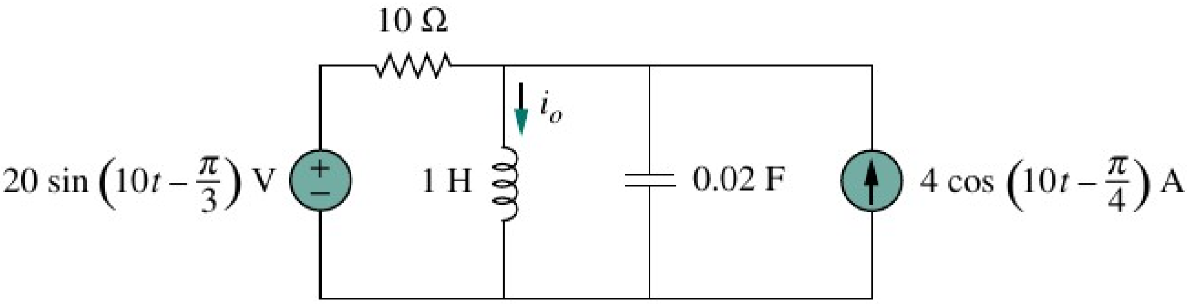


6) The switch in the circuit below şs moved from prosition a to b at time $t = 0$. Determine $i(t)$ for $t > 0$

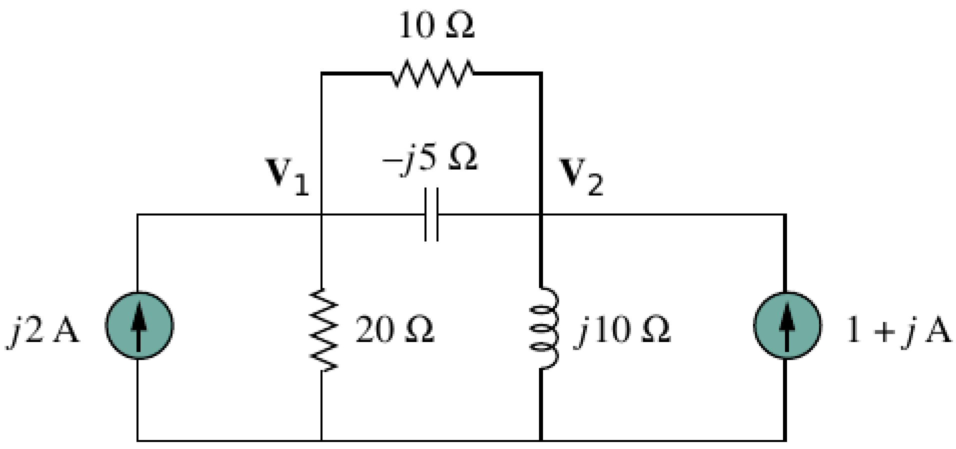


Sinosoidal Statedy State :

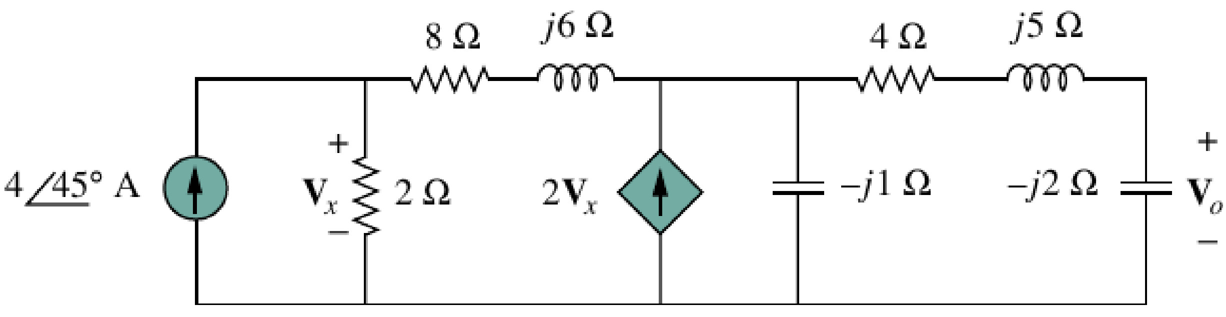
1) Given the following circuit determine $i_o(t)$



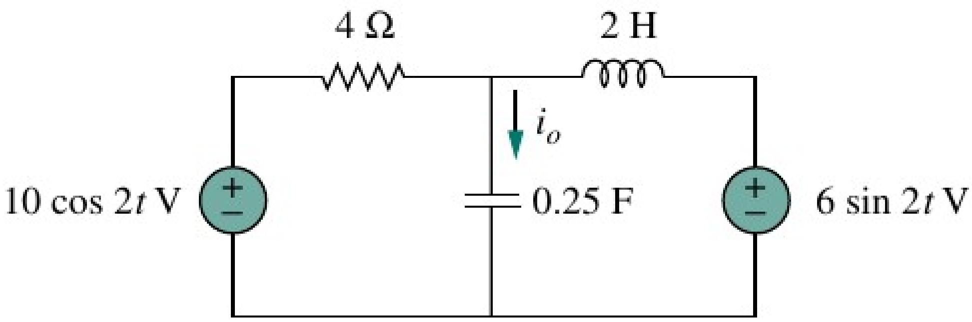
2) Using nodel analysisi find the values of V_1 and V_2 for the circuit given below



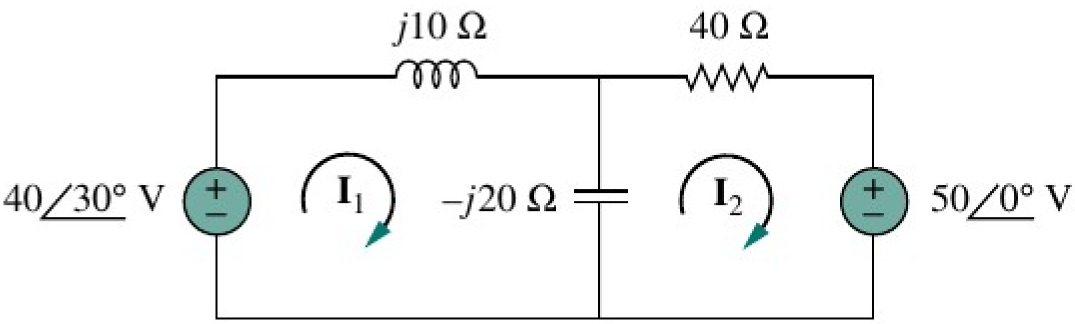
3) Obtain V_o using nodal analysis for the circuit given below



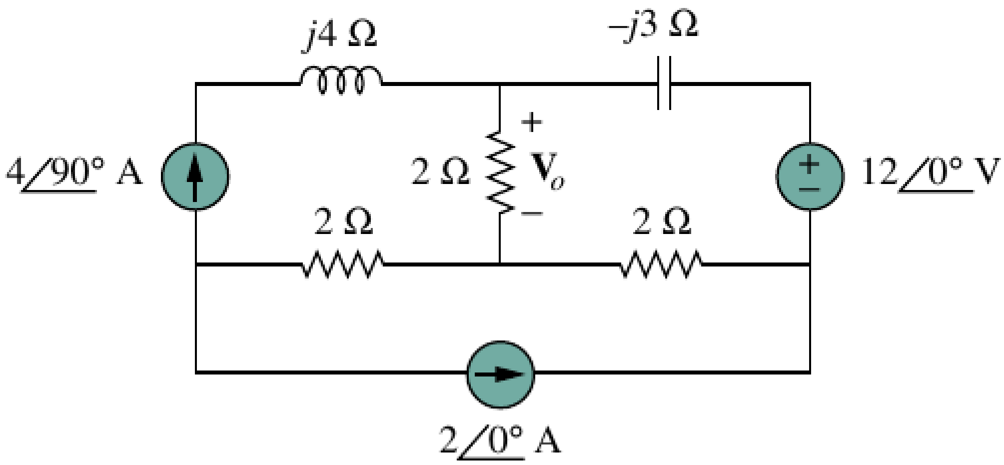
4) Solve for i_o for the given circuit using mesh analysisi



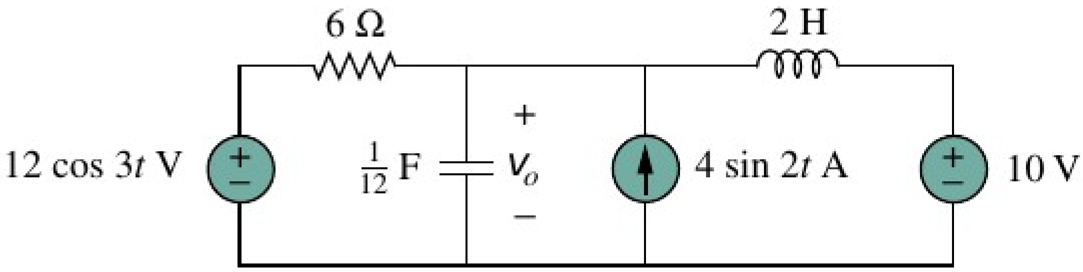
5) Using mesh analysis find I_1 and I_2 in the circuit depicted below



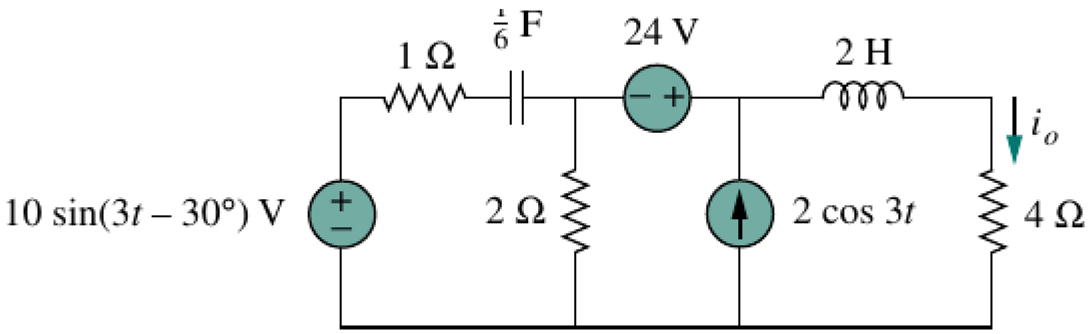
6) Compute V_o in the circuit given below using mesh analysisi



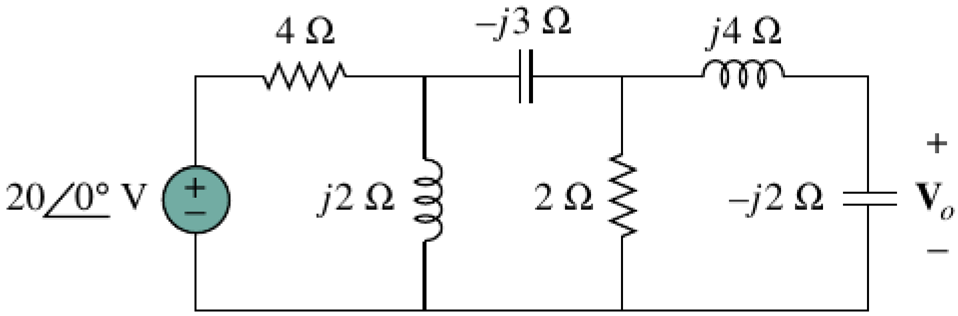
7) Solve for $v_o(t)$ for the given circuit (try it with superposition)



8) Compute $i_o(t)$ via source transformation



9) Via source transformation compute V_o for the circuit given below



10) Find the Thevenin equivalent of the circuit given below as seen from
   a) termşnals a-b
   b) terminals c-d